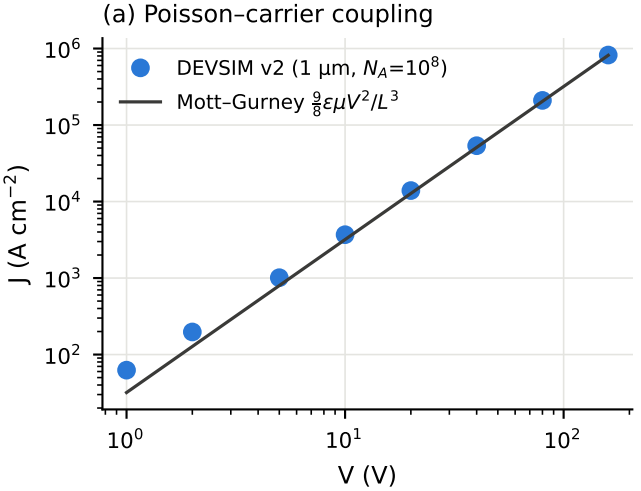
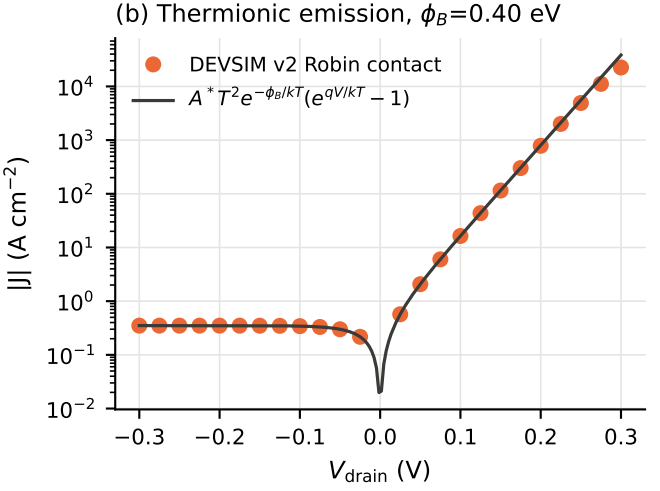

(a) Poisson–carrier coupling
DEVSIM v2 (1 μm, N_A=10^8)
Mott–Gurney $\frac{9}{8}\varepsilon\mu V^2/L^3$
J (A cm^{-2})
V (V)


(b) Thermionic emission, ϕ_B=0.40 eV
DEVSIM v2 Robin contact
$A^*T^2e^{-\phi_B/kT}(e^{qV/kT}-1)$
|J| (A cm^{-2})
V_{drain} (V)